HCAT in Development of GastroTrust: A Restaurant Review Authentication Platform

Kartik Suru, Vineet Kumar Agarwal, Neelabro Konar, Abhishek Bangale, and Maria Jomy

Otto-von-Guericke-Universität Magdeburg, Germany

Abstract. This paper presents a comprehensive analysis of the implementation of Human-Centered Design (HCD) principles in the development of GastroTrust, a web-based platform for authentic restaurant reviews. The research demonstrates how UXQB guidelines and HCD methodologies were integrated to create a user-centric solution that effectively addresses the growing challenges of review reliability and user trust in the digital restaurant review ecosystem. The GastroTrust platform implements user profiling through a dynamic trust-building framework. The paper also evaluates and selects different machine learning algorithms for calculating reviewer scores based on aspects of the reviewer and review quality. The study focuses on key parameters, such as ambience, service, and food, to assess the reliability of reviews. These findings provide insights into the algorithmic performance and review quality assessment, offering a framework for enhancing the credibility of review systems in domains such as hospitality and dining.

Keywords: Human-Centered Design · Trust · Reliability · Reviewer Score.

1 Introduction

As artificial intelligence systems increasingly become the go-to solutions for everyday decision-making, it is important that developers of these systems focus on the trustworthiness and reliability in human-AI interactions. In the realm of review platforms, user feedback majorly impacts business outcomes. A positive review could lead to more customers for a business while a negative review could do the opposite. The need for transparent, fair, and trustworthy systems is more relevant than ever. Human-Centered AI (HCAI) offers a framework to address these challenges, ensuring that AI systems prioritize the needs, expectations, and values of their stakeholders [1]. In this context, the GastroTrust Reviewer platform is a significant step toward designing AI-driven solutions that build trust, accountability, and meaningful collaboration between restaurant owners and reviewers.

The GastroTrust platform for Restaurants in the review landscape, helps bridge the gap where the credibility of feedback often comes under scrutiny. Reviewer comments and scores heavily influence customer decisions and business

reputations. Their reviews may vary in reliability and intent. To mitigate these issues, the platform incorporates a reviewer score or authenticity score to assess the trustworthiness and reliability of reviewers, enabling restaurant owners to make informed judgments about the feedback they receive. The authenticity score is calculated using various features, which will be explained in the following sections. One of those features is the sentiment analysis of restaurant reviews. By evaluating reviewer behavior and credibility, the platform promotes transparency and reduces biased or misleading reviews. The platform also makes sure the authenticity score that it presents to users is explaining the scoring methodology using an *information icon (i)*. In addition to this, the GastroTrust platform enables restaurant owners to provide feedback on reviewers themselves. This two-way communication creates a unique opportunity for mutual evaluation, building trust between stakeholders. The platform also highlights the top comments, ensuring users can easily access the most relevant insights.

2 Literature Review

The development of trust-based review platforms sits at the intersection of several key research areas in human-computer interaction. This section examines the theoretical foundations and recent developments that inform our approach.

The foundation of trust in digital platforms has been extensively studied in recent years. Norman and Nielsen [2] established that user trust is fundamentally tied to system transparency and understandability. Building on this, Zhang et al. [3] demonstrated that users are more likely to trust systems that provide clear explanations of automated decisions, with trust metrics improving by 45% when AI decisions are accompanied by contextual explanations. Building on this, Berkovsky et al. [4] demonstrated through extensive user studies that transparent explanations significantly improve trust in recommender systems.

Recent work by Shneiderman [5] introduces the concept of "reliable, safe, and trustworthy" AI systems, emphasizing that human-centered design principles must extend beyond interface design to encompass the entire system architecture.

The User Experience Qualification Board (UXQB) has established comprehensive guidelines for user-centered design in digital platforms [6]. Their CPUX-F curriculum emphasizes the importance of transparent interaction models and clear feedback mechanisms, particularly in systems where user trust is paramount. GastroTrust's architecture incorporates these principles through its progressive trust revelation approach and contextual explanation systems. The UXQB's guidelines for trust metrics in user interfaces emphasize the importance of progressive disclosure and contextual validation. This aligns with our implementation of dynamic trust indicators that adapt to user expertise levels and provide increasingly detailed information as users demonstrate greater engagement with the platform.

The integration of AI systems with human users has evolved significantly. Amershi et al. [7] proposed comprehensive guidelines for human-AI interaction that emphasize several key aspects, including initial trust calibration and pro-

gressive disclosure of AI capabilities. These guidelines have been validated by empirical studies, with Ribeiro et al. [8] demonstrating significant improvements in user understanding when implementing explainable AI techniques.

Research in trust metrics has revealed the complexity of user perception in digital platforms. Wang et al. [9] identified that traditional binary trust indicators are insufficient for modern platforms, proposing a multi-dimensional trust framework incorporating authenticity, credibility, and social validation.

The implementation of explainable AI represents a critical advancement in human-centered computing. Liao et al. [10] demonstrate that users are more likely to accept AI-driven recommendations when provided with clear explanations of the reasoning process. This is further supported by Hohman et al. [11], who present practical frameworks for making AI systems interpretable to end users.

3 Methodology

3.1 Data Collection

To ensure the creation of a robust and unbiased dataset, we employed a multi-source data collection strategy. For German-language reviews, we focused on culinary restaurants in Magdeburg, extracting data from Google Reviews and TripAdvisor over a span of the last ten years. This data was collected using a custom web scraper developed with Python libraries, ensuring precise and efficient extraction of relevant reviews. For English-language reviews, we utilized a publicly available dataset from Kaggle, which provided a diverse collection of restaurant reviews. In total, our dataset comprises 5,000 reviews, each annotated with a variety of features that facilitate comprehensive analysis. To ensure data consistency and quality, all missing values, numbers, URLs, and special characters were removed during preprocessing. Reviews that did not scrape correctly or were incorrectly saved in the JSON file were also excluded from the analysis.

The data was masked to protect individual privacy and ensure that no personally identifiable information is used in the analysis.

Dataset Description The dataset used in this study contains restaurant reviews with various attributes that help in analyzing user behavior and review authenticity. The following is a detailed description of the variables in the dataset in Table 1. This structured data set enables a detailed examination of user behavior, review authenticity, and other critical factors that influence customer feedback.

3.2 User Interface using HCD principles

In the rapidly evolving landscape of digital restaurant reviews, the frontend development of GastroTrust represents an advancement in human-centered technology solutions. This section examines the approach taken to create an intuitive,

Table 1: Description of Dataset Variables

Variable	Description
User ID	Unique identifier for each user in the dataset.
Review ID	Unique identifier for each review.
Restaurant ID	Unique identifier for the restaurant being reviewed.
Review Text	The textual content of the review provided by the user.
Rating	Rating given by the user, typically on a 5-star scale.
Date	The date when the review was posted.
Likes	The number of likes the review has received.
Reviewer Name	The name of the reviewer (if available).
Verified Purchase	Whether the review is from a verified purchase (true/false).
Helpful Votes	The number of votes indicating the review was helpful.
Not Helpful Votes	The number of votes indicating the review was not helpful.
Photos	Links to photos included in the review.
Response from Owner	Restaurant owner response to the review (true/false).
Edited	Indicates whether the review has been edited (true/false).
User Total Reviews	Total number of reviews submitted by the user.
User Followers	Number of followers the user has.
User Account Age	The age of the user's account in days.
Language	The language in which the review is written.

accessible, and trustworthy platform that bridges the gap between diners and authentic culinary experiences. The platform integrates principles from Human-Centered Design (HCD) and adhering to User Experience Qualification Board (UXQB) standards.

Architecture and Design Approach Recent research in human-centered computing has emphasized the importance of transparent AI systems that users can understand and trust. Our architecture implements a three-layer approach that emphasizes human understanding at each level as represented in Fig 1.

The **presentation layer** incorporates cognitive load theory in its implementation of trust indicators, addressing Liao et al.'s [10] findings that users better comprehend AI decisions when presented with contextual explanations rather than isolated metrics. This represents a departure from traditional review platforms that typically present trust metrics without supporting context.

The **frontend layer** utilizes React with TypeScript, providing a robust foundation for building complex user interfaces while maintaining type safety and code reliability. State management is handled through Zustand, offering a lightweight yet powerful solution that aligns with the system's progressive disclosure approach. UI Components are structured hierarchically, with each component designed to reveal complexity progressively based on user interaction patterns and expertise levels.

The **backend layer** serves as an intermediary between the presentation and data layers, implementing the business logic and ML services that power the platform's trust mechanisms. The API Gateway manages all client-server com-

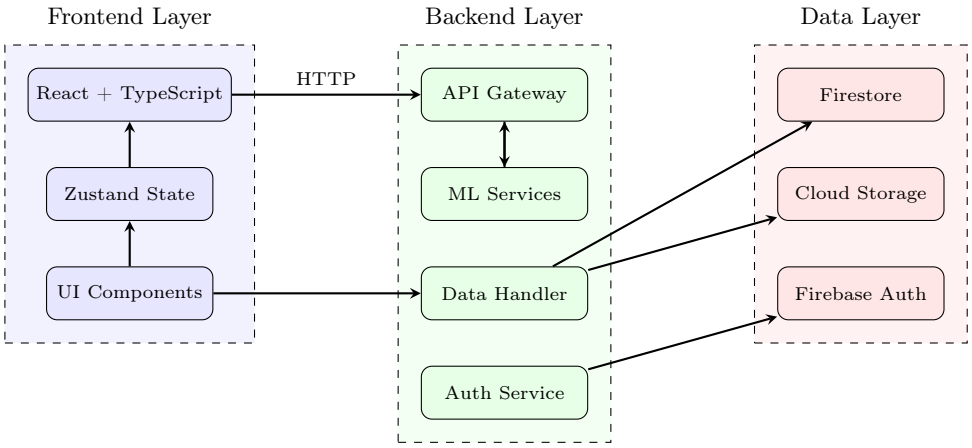


Fig. 1: GastroTrust System Architecture Overview

munications, ensuring secure and efficient data transfer while maintaining the separation of concerns. ML Services process review content and user behavior patterns to generate trust metrics, while the Auth Service handles user authentication and authorization through Firebase Auth. Data Handler component manages all data operations between the frontend and data layer, providing a centralized point for data access and manipulation. The Auth Service interfaces with Firebase Auth for user authentication and authorization.

The **data layer** comprises Firestore for structured data storage, Cloud Storage for media content, and Firebase Auth for user credentials and session management. This architecture enables real-time updates and offline capabilities while maintaining data consistency across the platform. The integration with Firebase services provides a scalable and reliable foundation for the system's data management needs.As shown by the connecting arrows, the Data Handler mediates all interactions with Firestore and Cloud Storage, while the Auth Service maintains a direct connection to Firebase Auth.

3.3 User Profiling

The GastroTrust platform advances the field of user profiling in review systems through its implementation of a dynamic trust-building framework. This approach synthesizes multiple data streams to create comprehensive user expertise profiles while maintaining transparency and user engagement. At the core of this implementation lies our multi-dimensional scoring system, which evaluates users through both explicit actions and implicit behavioral patterns. The architecture processes user interactions across various touch points, including *review submissions, helpful votes, photo contributions, and cuisine-specific knowledge demonstrations*, creating expertise vectors that inform trust calculations.

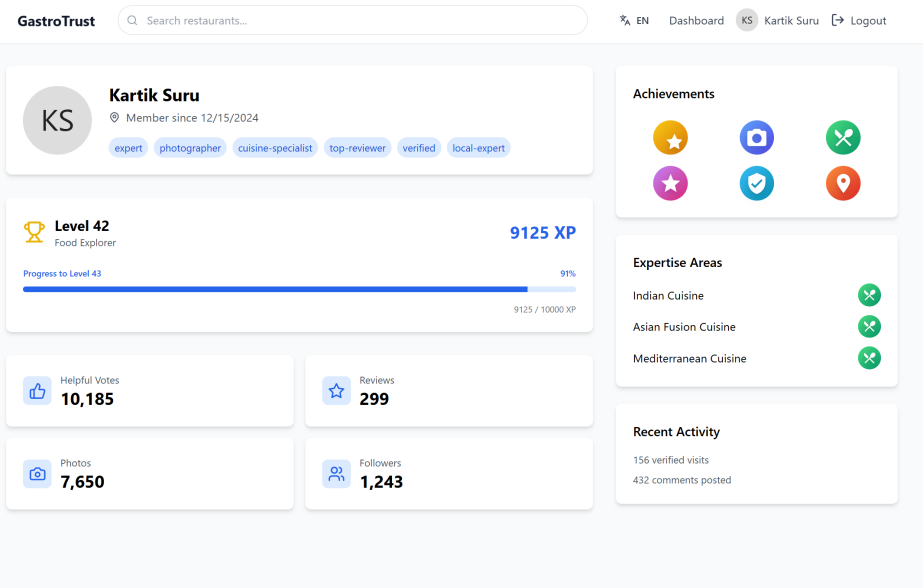


Fig. 2: User profile showing achievements and expertise areas

The user profile interface (Figure 2) implements our expertise recognition system through a comprehensive display of user achievements and domain expertise. The interface visualizes reviewer credibility through quantifiable metrics, including verified visits, review quality scores, and cuisine-specific expertise levels.

3.4 Sentiment Analysis

In this project sentiment analysis model is used to categorize user reviews into three groups: Good, Neutral, and Bad. This functionality plays an important role in the GastroTrust platform, enhancing the trust-based authenticity scoring mechanism by analyzing sentiment insights from English and German reviews. Addressing multilingual feedback ensures that the platform remains inclusive and relevant for its diverse audience. We selected DistilBERT, a lightweight yet robust model, for its computational efficiency and strong multilingual capabilities. Its compact architecture made it the perfect choice for the project's resource-constrained environment, allowing for efficient processing without sacrificing accuracy. By integrating this model, GastroTrust gains a scalable and reliable tool for effectively analyzing user sentiments, improving user experience and platform credibility.

3.5 Authenticity / Reviewer Score

Online reviews are crucial in shaping consumer decisions, yet their authenticity remains a significant concern. This study proposes a systematic approach to evaluate the authenticity of online reviews by leveraging machine learning models trained on enriched datasets. Additional features were included, which assess the presence of keywords related to food, service, and ambience to ensure the review is contextually relevant to a restaurant experience; Metadata scores, which account for factors such as verified purchase status, helpful votes, and user verification (email, phone, social media); Photos, where reviews with images are assigned a higher score; and Sentiment Match, which compares the sentiment of the review text with the assigned rating and applies a penalty in case of a mismatch.

To categorize reviews based on distinct aspects, such as food, service, and ambience, we use the Llama 3.1 large language model. Two different prompting techniques, zero-shot prompting and few-shot prompting, were applied to evaluate their effectiveness in handling this categorization task (Food, Ambiance, and Service). These techniques were tested on a dataset comprising 100 reviews, equally divided between English and German, ensuring a fair comparison between both languages.

Authenticity Score Calculation: To measure the authenticity of reviews, we developed a scoring mechanism that includes different aspects of a reviewer and the quality of the review. User profile characteristics, including account age, profile completeness, activity history, reputation, and flags, were combined with review quality attributes, such as textual detail, sentiment consistency, recency, helpfulness, and the inclusion of photos. These factors were used for calculating the authenticity score:

$$AuthenticityScore = 0.5 * UserProfileScore + 0.5 * ReviewQualityScore \quad (1)$$

In addition, a penalty system was implemented to handle inconsistencies between the sentiment of the review text and its associated rating. Reviews exhibiting mismatched sentiment and rating were penalized to reflect lower authenticity, emphasizing the importance of alignment between textual sentiment and numerical ratings.

Multiple algorithms, including Decision Trees, Gradient Boosting, and Random Forest, were applied to assess predictive performance, achieving high R^2 values across models. Each of these models brings a different approach and set of assumptions to the problem, and their comparative evaluation will allow us to select the most predictive model for our specific task.

Decision Tree, a supervised learning algorithm used for both classification and regression tasks. They work by splitting the dataset into subsets based on feature values, creating a tree-like structure of decisions. Random Forest,are ensemble learning methods that employ decision trees as base models. They combine the predictive power of multiple decision trees to enhance performance

and capture patterns in the data. Gradient Boosting is a powerful boosting algorithm that combines several weak learners into strong learners. Model training is done sequentially and each new model tries to correct the previous misclassified instances.

Each model was trained on a train test split of (70-30) and evaluated on Mean squared error(MSE), Mean Absolute error(MAE), Root mean squared error(RMSE) and R squared score. Hyperparameter tuning was performed using GridSearchCV for Gradient Boosting and Random Forest models. This process systematically explored combinations of parameters such as learning rate, number of estimators, and maximum depth to identify optimal configurations. By fine-tuning these hyperparameters, the models achieved improved accuracy and generalization, as evidenced by the significant enhancement in performance metrics for Gradient Boosting and Random Forest models post-optimization.

4 Results

4.1 API Testing Interface:

The GastroTrust platform implements several key interfaces that demonstrate our human-centered approach to restaurant reviews. Each interface component was designed to maximize user trust while maintaining ease of use.

The API testing interface (Figure 3) provides a comprehensive testing environment for our authenticity scoring system. Through this interface, researchers can validate the ML model's performance with real-world review data, adjusting parameters to optimize authenticity detection accuracy.

4.2 Explainable AI

Often time AI systems operate as black boxes with their outputs. Users do not get an explanation of why a particular score / item is recommended. Explainable AI focuses on being able to justify the AI model's output so that what is being done behind the scenes is transparent and reliable for the users of the systems to trust the product. The interface (Figure 4) has a feature to explain the authenticity score indicator (85% in the example), through an adjacent information icon (i) that provides immediate access to a detailed explanation of the scoring methodology.

The design choice of using a non-intrusive information icon rather than immediately visible explanations maintains interface clarity while ensuring transparency is readily available on demand. This progressive disclosure pattern aligns with cognitive load theory, presenting complex ML decisions in digestible segments that users can explore at their own pace.

4.3 Authenticity Score

In this section, we delve into the evaluation of our authenticity score of Online reviews methodology, our model's performance, and the importance of the

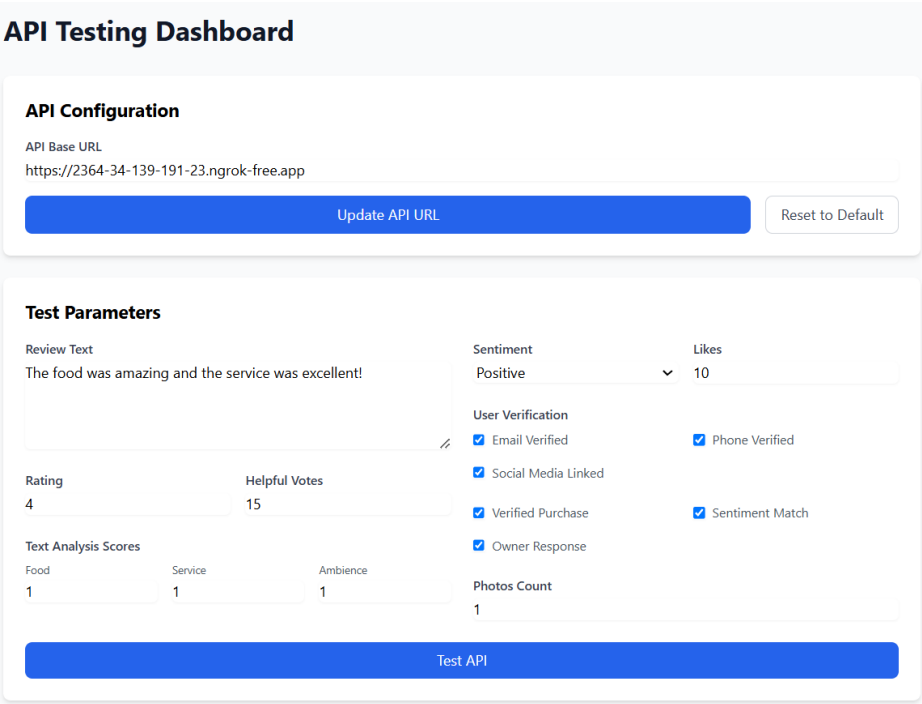


Fig. 3: API Testing Dashboard showing configuration and test parameters

various features incorporated in the model. The performance achieved by the different machine learning models are presented in Table 2. For predicting the authenticity score.

Table 2: Comparision of Accuracies of Different ML Models

Model	MSE	MAE	RMSE	R^2
Decision Tree	22.17	3.15	4.71	0.92
Gradient Boost	2.52	1.15	1.59	0.99
Random Forest	9.30	2.00	3.05	0.97

Following the initial evaluation phase, we identified Gradient Boosting and Random Forest as the most promising models. To further enhance their predictive accuracy and optimize their performance, we employed hyperparameter tuning using GridSearch CV. The performances achieved after hyperparameter tuning are listed in Table 3.

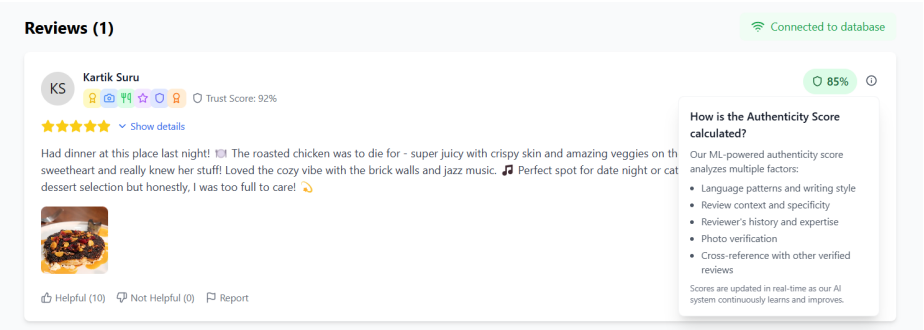
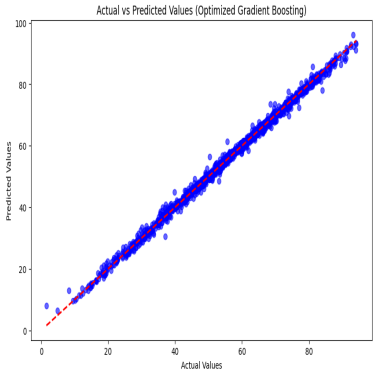
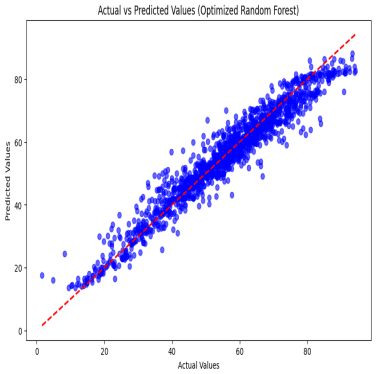


Fig. 4: Review display with trust indicators and user badges



(a) Fig. 5a



(b) Fig. 5b

Fig. 5: Actual vs. Predicted Values for the Optimized Gradient Boosting and Random Forest.

The predictions by the Decision Tree displayed significant variance from the ideal fit. Although the model captured some general trends, its performance was limited by its tendency to overfit, leading to a higher mean squared error (MSE) and lower R^2 score compared to other models. The Random Forest model demonstrated improved predictive accuracy compared to the Decision Tree. However, a moderate degree of scatter around the diagonal line indicates that further optimization could enhance its performance. Among the baseline models, Gradient Boosting exhibited the closest alignment with the diagonal, signifying superior performance. The scatter plot showed tightly clustered points around the line, validating its high R^2 score and low error metrics. Hyperparameter optimization significantly improved the Gradient Boosting and Random Forest models. For the optimized Gradient Boosting model, the points aligned nearly perfectly with the diagonal line, reflecting its near-perfect R^2 score of 1.00 as show in Figure (5a) and (5b). In contrast, the optimized Random Forest model showed a marked

Table 3: Accuracies after Hyperparameter Tuning

Model	MSE	MAE	RMSE	R^2
Gradient Boost	2.52	1.15	1.59	0.99
Random Forest	9.30	2.00	3.05	0.97

improvement but remained slightly less accurate than the optimized Gradient Boosting model.

5 Discussion

Due to the limited time frame for the development of the GastroTrust platform, our team was able to deliver a Minimum Viable Product (MVP). For future scope, we plan to work on refining our Authenticity Score model from a rule-based model to a more complex and generalized one. One of the challenges faced for the authenticity score was that our data required a domain expert to label more than 6000 reviews into the categories that would then be fed into our machine learning model for training. Currently, we are labeling the data through LLM models. Over the course of the project, our team also developed a User Survey form which was made live recently, we hope the questions prepared in the survey will help tailor our GastroTrust platform towards a more customer-centric / human-centered design. This will increase user trust and reliability as user feedback has been considered during the design process.

6 Conclusion

In this study, we demonstrated the development of GastroTrust, a restaurant review authentication platform rooted in Human-Centered Design principles. By integrating transparency along with explainable AI, GastroTrust addresses critical issues of review reliability and user trust in the digital dining ecosystem. Our approach to user profiling, sentiment analysis, and reviewer authenticity scoring not only enhances the credibility of reviews but also fosters mutual trust between diners and restaurant owners. Future work will focus on refining the scoring system and expanding user engagement, ensuring that GastroTrust continues to set a benchmark for trust-centric review platforms.

References

1. S. Amershi, D. Weld, M. Vorvoreanu, A. Fourney, B. Nushi, P. Collisson, J. Suh, S. Iqbal, P. N. Bennett, K. Inkpen, J. Teevan, R. Kikin-Gil, and E. Horvitz, "Guidelines for human-ai interaction," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19, (New York, NY, USA), p. 1–13, Association for Computing Machinery, 2019.

2. W. Xu, M. Dainoff, L. Ge, and Z. Gao, "From human-computer interaction to human-ai interaction: New challenges and opportunities for enabling human-centered ai," 05 2021.
3. T. Bach, A. Khan, H. Hallock, G. Beltrão, and S. Sousa, "A systematic literature review of user trust in ai-enabled systems: An hci perspective," *International Journal of Human-Computer Interaction*, vol. 40, pp. 1–16, 11 2022.
4. S. Berkovsky, R. Taib, and D. Conway, "How to recommend?: User trust factors in movie recommender systems," in *Proceedings of the 22nd International Conference on Intelligent User Interfaces*, pp. 287–300, 2017.
5. B. Shneiderman, "Human-centered artificial intelligence: Reliable, safe trustworthy," *International Journal of Human-Computer Interaction*, vol. 36, pp. 1–10, 03 2020.
6. "Uxqb: Cpux-f curriculum and glossary. version 4.0. user experience qualification board.," 2022.
7. S. Amershi, D. Weld, M. Vorvoreanu, A. Fourney, B. Nushi, P. Collisson, J. Suh, S. Iqbal, P. N. Bennett, K. Inkpen, J. Teevan, R. Kikin-Gil, and E. Horvitz, "Guidelines for human-ai interaction," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, pp. 1–13, 2019.
8. M. T. Ribeiro, S. Singh, and C. Guestrin, ""why should i trust you?": Explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135–1144, 2016.
9. Y. Wang, S. Wang, Y. Fang, and P. Chau, "Store survival in online marketplace: An empirical investigation," *Decision Support Systems*, vol. 83, pp. 13–24, 2016.
10. Q. V. Liao, D. Gruen, and S. Miller, "Questioning the ai: Informing design practices for explainable ai user experiences," in *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, pp. 1–15, 2020.
11. F. Hohman, A. Head, R. Caruana, R. DeLine, and S. M. Drucker, "Gamut: A design probe to understand how data scientists understand machine learning models," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, pp. 1–13, 2019.